THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Red litmus turns ___ in a basic solution. 1

(a) pink **(b)** yellow **(c)** blue **(d)** red

2. Which of these tastes bitter and feels soapy? 1

(a) Neutral salts **(b)** Acids **(c)** Sugars **(d)** Bases

3. The electricity from the power plant at home is: 1

(a) solar current **(b)** alternating current (AC)
(c) direct current only **(d)** static charge

4. Which of the following would NOT make a bulb glow in the circuit? 1

(a) An aluminium strip **(b)** A rubber band as connecting wire
(c) A copper wire **(d)** An iron nail

5. Metals are generally: 1

(a) poor conductors **(b)** brittle
(c) good conductors of heat and electricity **(d)** insulators

6. Burning magnesium ribbon gives: 1

(a) rust **(b)** a white powder
(c) a blue flame only **(d)** a black powder

7. Burning of paper is a: 1

(a) dissolving **(b)** physical change **(c)** chemical change **(d)** melting

8. Fermentation of grapes into wine is: 1

(a) cooling **(b)** a physical change
(c) a chemical change **(d)** dissolving

9. The growth spurt in height during adolescence is seen in: 1

(a) both boys and girls **(b)** girls only
(c) boys only **(d)** neither

10. Changes in boys and girls begin at the same age? 1

(a) Yes, at birth
(b) Only boys change
(c) Yes, exactly at 12
(d) No, the timing varies from person to person

11. In convection, warm fluid: 1

(a) sinks and cooler rises **(b)** rises and cooler fluid sinks
(c) stays still **(d)** moves sideways only

12. The normal temperature of the human body is about: 1

(a) 25 °C **(b)** 98 °C **(c)** 37 °C **(d)** 50 °C

13. A bus covers 180 km in 2 hours. Its average speed is: 1

(a) 90 km/h **(b)** 45 km/h **(c)** 360 km/h **(d)** 100 km/h

14. 15 m/s is equal to: 1

(a) 4.2 km/h **(b)** 54 km/h **(c)** 900 km/h **(d)** 150 km/h

15. Celiac disease is a reaction to: 1

(a) gluten **(b)** lactose **(c)** vitamin C **(d)** iron

16. Saliva helps to break down: 1

(a) fats **(b)** starch **(c)** proteins **(d)** fibre

17. The raw materials for photosynthesis are: 1

(a) glucose and oxygen **(b)** carbon dioxide and water
(c) oxygen and water **(d)** nitrogen and water

18. Which plant is a parasite? 1

(a) Cuscuta (amarbel) **(b)** Rose
(c) Mango **(d)** Hibiscus

19. A pinhole camera forms an image that is: 1

(a) inverted **(b)** erect
(c) coloured only **(d)** bigger than object always

20. In summer in the northern hemisphere, the North Pole is tilted: 1

(a) away from the Sun **(b)** towards the Sun
(c) does not tilt **(d)** sideways only

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 15–25 words each.

21. A solution does not change litmus colour. What type of solution is it? Give an example. 2

22. State whether a graphite pencil lead conducts. Why is that unusual? 2

23. Why are cooking vessels made of metals? 2

24. Name a government initiative for menstrual hygiene. 2

25. Why does a land breeze blow at night? 2

26. Express 5 hours 30 minutes in minutes. 2

27. Name one harmful effect of smoking. 2

28. What are saprotrophs? Give an example. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. What is the difference between a conductor and an insulator in terms of use in cables? 3

30. Why is the use of plastic bags considered both useful and undesirable? 3

31. Why does condensation form on a glass of cold water? 3

32. Why are lungs protected by the rib cage? 3

33. Why can we not see around a corner directly? 3

34. Why does a total solar eclipse last only a few minutes at one place? 3

35. What happens when a drop of an acid is added to a base solution containing a few drops of indicator? Explain gradual change. 3

OR

Which is more suitable to relieve a stomach ache: lemon juice or milk of magnesia? Why? 3

36. Why is it harmful to keep too many plants in a closed bedroom at night? 3

OR

Why is Cuscuta called a parasite? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–120 words each.

37. (i) What is an alloy? 5
(ii) Give three alloys with their components.
(iii) Why are alloys useful?

OR

(i) Describe malleability and ductility. 5
(ii) Name two metals for each.
(iii) Which is the most malleable?

38. (i) Why is a ‘balanced social life’ necessary? 5
(ii) How to use technology responsibly?
(iii) Give two examples.

OR

(i) State the roles of RKSK, MHS and Suvidha initiatives. 5
(ii) Why are they important?

39. (i) Draw (describe) a distance–time graph for a bus stopping at a station. 5
(ii) Explain each part.

OR

(i) What is motion? 5
(ii) Types of motion with examples.
(iii) How is speed measured in vehicles?

40. (i) A boy stands 2 m from a plane mirror. How far is his image from him? 5
(ii) He moves 1 m closer. New distance?
(iii) State why.

OR

(i) Why does light need no medium? 5
(ii) How does Sun’s light reach us?
(iii) Compare with sound.

END OF QUESTION PAPER — Total: 80 Marks

INTERMIND